

WENTWORTH LIU

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Data Scientist · Applied ML · Statistical Modelling · Decision Science

SUMMARY

Data Scientist with 2+ years of experience applying Python, SQL, statistical modelling, and machine learning to complex operational and strategic decisions. Combines an Applied Science research background with commercial experience in explainable risk ranking, quasi-causal analysis, NLP, and reusable data products for million-scale EV fleets at Li Auto; previously delivered multi-source measurement and decision analytics at ByteDance. Strong at structuring ambiguous problems and translating quantitative findings into actionable recommendations.

EXPERIENCE

Li Auto

Data Scientist, Battery Intelligence Department

Beijing, China

Jul 2024 – Present

- **Statistical Modelling & Decision Support.** Used quasi-causal analysis on million-scale fleet telemetry to identify high-risk SOH-degradation cohorts and actionable drivers; productised intervention analysis across multiple vehicle models, supporting targeted strategies projected to reduce warranty exposure by tens of millions of RMB for a single model.
- **Risk Modelling & Operational Prioritisation.** Recast thermal-runaway detection under extreme class imbalance as an explainable ranking task; built a cross-platform ML pipeline with vehicle-level risk scores and feature-level drivers, achieving 93% Recall@2% on retrospectively validated high-risk cases while limiting inspection to the highest-risk 2% of vehicles.
- **Reusable Analytical Workflows.** Built a configurable feature system using custom event definitions, multi-operator aggregations, and cross-platform normalisation; deployed reusable workflows to an internal ML platform and improved feature-iteration efficiency by ~85%.
- **Applied NLP & Process Optimisation.** Built a hybrid rule-based and statistical classifier for battery-quality work orders across 30+ fault categories, combining text matching, TF-IDF, and Random Forest; achieved Weighted F1 > 0.90 and used confidence-based routing to focus review effort on ambiguous cases.

ByteDance

Data Analytics Intern, Douyin Market Strategy

Beijing, China

May – Sep 2023

- **Multi-source Measurement & Monitoring.** Integrated consumption, dissemination, and search signals across multiple data platforms into an IP popularity measurement system; added automated pipeline monitoring to address upstream data instability and support recurring content-investment and promotional decisions.
- **Segmentation & Strategic Analysis.** Redesigned performance metrics and built a multi-dimensional content taxonomy for Qishui Music; diagnosed new-user conversion decline through funnel segmentation and KOL audience-saturation modelling, informing content selection and marketing-resource allocation.

Dongwu Securities

Equity Research Intern, Mechanical Sector

China

Dec 2021 – Mar 2022

- **Quantitative Market Analysis.** Combined company fundamentals, competitive analysis, and market dynamics with stock-price modelling to deliver short-, medium-, and long-term investment recommendations for Harmonic Drive Systems.

EDUCATION

Nanjing University

MSc Computational Materials Science

Sep 2021 – Jun 2024

- Developed Skeaf, a 3D Fermi-surface visualisation tool combining HDBSCAN clustering with quantum-oscillation data; first author, Physical Review B (2024); co-author, Physical Review B (2024) and Physical Review Materials (2023); Patent: CN115795266A.

Nanjing University

BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort

Sep 2017 – Jul 2021

- National Scholarship (top 0.15%) · MCM/ICM 2020 Honorable Mention (Team Captain) · Sakura Science Exchange Programme, Tohoku University, Japan (5 selected university-wide).

SKILLS

Programming

Python · SQL/HiveSQL · C++ · MATLAB · Git

ML & Statistics

Machine Learning · Statistical Modelling · Predictive Modelling · Causal Inference · Decision Modelling · Model Evaluation · Feature Engineering · NLP · Time Series · LightGBM · scikit-learn · PyTorch

Data & Delivery

Multi-source Data Integration · Data Pipelines · Monitoring · Data Productisation · Operational Analytics · Visualisation · GCP · Docker · CI/CD · Spark · Hadoop · Linux

Domain & Lang.

Applied Science · Battery Intelligence · EV Analytics · Product Analytics · English (TOEFL 104) · Mandarin/Cantonese (Native) · Japanese (JLPT N1)